Lie Algebras

Labix

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Abstract

Potentially good books: Humphreys, Erdmann and Wildson

Contents

1 Introduction to Lie Algebras

1.1 Lie Algebras over a Commutative Ring

Definition 1.1.1 (Lie Brackets)
Let R be a commutative ring. Let M be an R-module. A Lie bracket of M is an R-bilinear map $[-,-] : M \times M \to M$ such that the following are true.

- $[x,x] = 0$ for all $x \in M$.
- $[[x,y],z] + [[y,z],x] + [[z,x],y] = 0$ for all $x,y,z \in M$.

Lemma 1.1.2
Let R be a commutative ring. Let M be an R-module. Let $[-,-] : M \times M \to M$ be a Lie bracket. Then we have

$$[x,y] = -[y,x]$$

for all $x,y \in M$.

Definition 1.1.3 (Lie Algebras)
Let R be a commutative ring. A Lie algebra over R is an R-module L together with a Lie bracket $[-,-] : L \times L \to L$.

1.2 Lie Subalgebras and Ideals

Definition 1.2.1 (Lie Subalgebras)
Let R be a commutative ring. Let L be a Lie algebra over R. Let $A \subseteq L$ be a subset. We say that A is a Lie subalgebra of L if the following are true.

- A is an R-submodule of L.
- $[a_1, a_2] \in A$ for all $a_1, a_2 \in A$.

Definition 1.2.2 (Ideals of a Lie Algebra)
Let R be a commutative ring. Let L be a Lie algebra over R. Let $I \leq L$ be an R-submodule of L. We say that I is an ideal of L if the following are true.

- I is an R-submodule of L.
- $[r,i] \in I$ for all $r \in R$ and $i \in I$.

Proposition 1.2.3
Let R be a commutative ring. Let L be a Lie algebra over R. Let I, J be ideals of L. Then the following are also ideals of L.

- The intersection $I \cap J$.
- The sum $I + J = \{i + j \mid i \in I, j \in J\}$.

Definition 1.2.4 (The Lie Bracket of Ideals)
Let R be a commutative ring. Let L be a Lie algebra over R. Let I, J be ideals of L. Define the Lie bracket of ideals of I and J to be

$$[I,J] = R\langle [i,j] \mid i \in I, j \in J \rangle$$

Lemma 1.2.5
Let R be a commutative ring. Let L be a Lie algebra over R. Let I, J be ideals of L. Then $[I,J]$ is an ideal of L.

1.3 Products and Quotients

Definition 1.3.1 (Direct Sum of Lie Algebras)
Let R be a commutative ring. Let L_1, L_2 be Lie algebras over R. Define the direct sum of L_1 and L_2 to be the R-module $L_1 \oplus L_2$ together with component wise Lie bracket.

Proposition 1.3.2
Let R be a commutative ring. Let L_1, L_2 be Lie algebras over R. Then the following are true.
- $[L_1 \oplus L_2, L_1 \oplus L_2] = [L_1 \oplus L_1] \oplus [L_2 \oplus L_2]$.
- $L_1 \oplus \{0\}$ is an ideal of $L_1 \oplus L_2$.
- $\{0\} \oplus L_2$ is an ideal of $L_1 \oplus L_2$.

Definition 1.3.3 (Quotient Lie Algebra)
Let R be a commutative ring. Let L be a Lie algebra over R. Let I be an ideal of L. Define the quotient Lie algebra to be the R-module

$$L/I = \{x + I \mid x \in L\}$$

together with Lie bracket $[x + I, y + I] = [x, y] + I$.

1.4 The Centers and Centralizers of Lie Algebras

Definition 1.4.1 (Center of a Lie Algebra)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the center of L by

$$Z(L) = \{z \in L \mid [z, x] = 0 \text{ for all } x \in L\}$$

Lemma 1.4.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then $Z(L)$ is an ideal of L.

Proposition 1.4.3
Let R be a commutative ring. Let L_1, L_2 be Lie algebras over R. Then we have

$$Z(L_1 \oplus L_2) = Z(L_1) \oplus Z(L_2)$$

Definition 1.4.4 (The Centralizer of a Subset)
Let R be a commutative ring. Let L be a Lie algebra over R. Let $A \subseteq L$ be a subset. Define the centralizer of A in L to be the set

$$C_L(A) = \{x \in L \mid [x, a] = 0 \text{ for all } a \in A\}$$

Lemma 1.4.5
Let R be a commutative ring. Let L be a Lie algebra over R. Let $A \subseteq L$ be a subset. Then $C_L(A)$ is a Lie subalgebra of L.

1.5 The Lie Algebra of Endomorphisms

Definition 1.5.1 (The Lie Algebra of Endomorphisms)
Let R be a commutative ring. Let L be a Lie algebra over R. Let R be a commutative ring. Let L be a Lie algebra over R. Define the Lie algebra of endomorphisms of L to be the R-module

$$\mathfrak{gl}_R(L) = \text{End}_R(L) = \{T : L \to L \mid T \text{ is } R\text{-linear }\}$$

together with Lie bracket $[-,-] : \mathfrak{gl}_R(L) \to \mathfrak{gl}_R(L)$ given by

$$[T, S] = T \circ S - S \circ T$$

A priori one needs to check that the above map is indeed a Lie bracket. Let $A \in \text{End}_R(L)$. Then $[A, A] = A^2 - A^2 = 0$ so that the alternating property is satisfied. For the Jacobi identity, we have that

$$\begin{aligned}
[[A,B],C] + [[B,C],A] + [[C,A],B] &= [A,B]C - C[A,B] + [B,C]A - A[B,C] + [C,A]B - B[C,A] \\
&= ABC - BAC - CAB + CBA + BCA - CBA \\
&\quad - ABC + ACB + CAB - ACB - BCA + BAC \\
&= 0
\end{aligned}$$

for all $A, B, C \in \text{End}_R(L)$.

Definition 1.5.2 (The Lie Algebra of the Matrix Ring)
Let R be a commutative ring. Define the Lie algebra of the matrix ring to be the R-module

$$\mathfrak{gl}_n(R) = M_n(R)$$

together with the Lie bracket $[-,-] : \mathfrak{gl}_n(R) \to \mathfrak{gl}_n(R)$ defined by

$$[A, B] = AB - BA$$

2 Lie Algebra Homomorphisms

2.1 Lie Algebra Homomorphisms

Definition 2.1.1 (Homomorphism of Lie algebra)
Let R be a commutative ring. Let L, M be Lie algebras over R. Let $\varphi : L \to M$ be an R-module homomorphism. We say that φ is a Lie algebra homomorphism if

$$[\varphi(a), \varphi(b)] = \varphi([a, b])$$

for all $a, b \in L$.

Definition 2.1.2 (Kernel of a Lie Algebra Homomorphism)
Let R be a commutative ring. Let L, M be Lie algebras over R. Let $\varphi : L \to M$ be an R-module homomorphism. Define the kernel of φ to be

$$\ker(\varphi) = \{x \in L \mid \varphi(x) = 0\}$$

Lemma 2.1.3
Let R be a commutative ring. Let L, M be Lie algebras over R. Let $\varphi : L \to M$ be a Lie algebra homomorphism. Then $\ker(\varphi)$ is a Lie subalgebra of L.

Proof It is clear that $\ker(F)$ is a vector subspace of V. Let $k_1, k_2 \in \ker(F)$. Then we have

$$\begin{aligned} F([k_1, k_2]) &= [F(k_1), F(k_2)] && (F \text{ is a Lie algebra homomorphism}) \\ &= [0, 0] \\ &= 0 \end{aligned}$$

Hence $[k_1, k_2] \in \ker(F)$ and $\ker(F)$ is a Lie subalgebra of V. ∎

Definition 2.1.4 (Isomorphisms of Lie Algebras)
Let R be a commutative ring. Let L, M be Lie algebras over R. Let $\varphi : L \to M$ be a Lie algebra homomorphism. We say that φ is an isomorphism if there exists a Lie algebra homomorphism $\psi : M \to L$ such that

$$\psi \circ \varphi = \mathrm{id}_L \quad \text{and} \quad \varphi \circ \psi = \mathrm{id}_M$$

Theorem 2.1.5 (First Isomorphism Theorem)
Let R be a commutative ring. Let L, M be Lie algebras over R. Then the following are true.

- $\ker(\phi)$ is an ideal of L_1
- $\mathrm{im}(\phi)$ is a Lie subalgebra of L_2

Moreover, we have an isomorphism

$$\frac{L_1}{\ker(\phi)} \cong \mathrm{im}(\phi)$$

Theorem 2.1.6 (Second Isomorphism Theorem)
Let R be a commutative ring. Let L be a Lie algebra over R. Let I, J be ideals of R. Then the following are true.

- I and J are ideals of $I + J$
- $I \cap J$ is an ideal of I and J

Moreover, we have an isomorphism

$$\frac{I + J}{J} \cong \frac{I}{I \cap J}$$

Theorem 2.1.7 (Third Isomorphism Theorem)
Let R be a commutative ring. Let L be a Lie algebra over R. Let I and J be ideals of L such that

$I \subseteq J$. Then J/I is an ideal of L/I. Moreover, there is an isomorphism

$$\frac{L/I}{J/I} \cong \frac{L}{J}$$

Theorem 2.1.8 (Correspondence Theorem)
Let R be a commutative ring. Let L be a Lie algebra over R. Let I be an ideal of R. Then there exists a bijective correspondence

$$\{J \mid J \text{ is an ideal of } L \text{ and } I \subseteq J\} \quad \stackrel{1:1}{\longleftrightarrow} \quad \{K \mid K \text{ is an ideal of } L/I\}$$

2.2 The Adjoint Homomorphism

Definition 2.2.1 (The Adjoint Homomorphism)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the adjoint homomorphism $\text{ad} : L \to \text{End}_R(L)$ to be the map given by

$$\text{ad}(x)(y) = [x, y]$$

Lemma 2.2.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then then the adjoint homomorphism $\text{ad} : L \to \text{End}(L)$ is a Lie algebra homomorphism.

Proof

- Linearity: Let $x, y \in L$ and let $a, b \in R$. For any $z \in L$, we have

$$\begin{aligned} \text{ad}(ax + by)(z) &= [ax + by, z] \\ &= a[x, z] + b[y, z] \\ &= a\text{ad}(x)(z) + b\text{ad}(y)(z) \\ &= (a\text{ad}(x) + b\text{ad}(y))(z) \end{aligned}$$

  so that $\text{ad} : L \to \text{End}_R(L)$ is a linear map.
- Preserving the Lie bracket: Let $x, y \in L$. For any $z \in L$, we have

$$\begin{aligned} [\text{ad}(x), \text{ad}(y)](z) &= (\text{ad}(x)\text{ad}(y) - \text{ad}(y)\text{ad}(x))(z) \\ &= \text{ad}(x)(\text{ad}(y)(z)) - \text{ad}(y)(\text{ad}(x)(z)) \\ &= \text{ad}(x)([y, z]) - \text{ad}(y)([x, z]) \\ &= [x, [y, z]] - [y, [x, z]] \\ &= -[[y, z], x] + [[x, z], y] \\ &= -[[y, z], x] - [[z, y], x] - [[y, x], z] \\ &= -[[y, z], x] + [[y, z], x] - [[y, x], z] \\ &= [[x, y], z] \\ &= \text{ad}([x, y])(z) \end{aligned}$$

  Thus we have showed that $[\text{ad}(x), \text{ad}(y)] = \text{ad}([x, y])$.

■

Lemma 2.2.3
Let R be a commutative ring. Let L be a Lie algebra over R. Then we have

$$\ker(\text{ad}) = Z(V)$$

Proof Let $k \in \ker(\text{ad})$. Let $v \in L$. Then $[k, v] = \text{ad}(k)(v) = 0$ since $\text{ad}(k) = 0 \in \text{End}(L)$. Hence $k \in Z(L)$. Conversely, if $z \in Z(L)$ then we have

$$\text{ad}(z)(v) = [z, v] = 0$$

for all $v \in L$. Hence $\text{ad}(z) = 0 \in \text{End}_R(L)$ and $z \in \ker(\text{ad})$. ■

2.3 Ad-Nilpotency

Definition 2.3.1 (Ad-Nilpotency)
Let L be a Lie algebra. Let $x \in L$. We say that x is ad-nilpotent if $\text{ad}(x)$ is a nilpotent in $\mathfrak{gl}(V)$. (as an element of a ring).

Lemma 2.3.2
Let V be a vector space. Let $L \subseteq \text{End}(V)$ be a Lie subalgebra. If $x \in L$ is nilpotent, then x is ad-nilpotent.

Let V be a vector space over a field k. Let $T \in \text{End}(V)$. Recall from Linear Algebra that a Jordan-Chevalley decomposition is two linear maps $D, S \in \text{End}(V)$ such that the following are true:

- $T = D + S$
- D is diagonal and S is nilpotent.
- $DS = SD$

We showed that such a decomposition always exists and is unique.

Proposition 2.3.3
Let V be a finite dimensional vector space over a field k. Let $T \in \text{End}_k(V)$. Let $T = D + S$ be the unique Jordan-Chevalley decomposition. Then

$$\text{ad}(T) = \text{ad}(D) + \text{ad}(S)$$

is the Jordan-Chevalley decomposition of $\text{ad}(T) \in \text{End}(\text{End}(V))$.

Proof Let $T \in \text{End}_k(V)$ be an endomorphism. Let $T = D + S$ be the Jordan-Chevalley decomposition of T. Since ad is linear, we have that

$$\text{ad}(T) = \text{ad}(D) + \text{ad}(S)$$

For any $C \in \text{End}_k(V)$, $\text{ad}(D)$ is defined by $\text{ad}(D)(C) = DC - CD$. Since D is diagonalizable, we can choose a basis $B = \{b_1, \dots, b_n\}$ of V such that the matrix representing D given by $D_B = \text{diag}(\alpha_1, \dots, \alpha_r)$ is diagonal on the basis. For the standard basis $\{e_{i,j} \mid 1 \leq i, j \leq n\}$ on $\text{End}_k(V)$, we have that

$$\text{ad}(D)(e_{i,j}) = [D_B, e_{i,j}] = (\alpha_i - \alpha_j)e_{i,j}$$

which shows that every standard basis vector is an eigenvector of $\text{ad}(D)$. Hence $\text{ad}(D)$ is diagonal. On the other hand, since S is nilpotent, by the above $\text{ad}(S)$ is nilpotent.

Finally, we have that

$$(\text{ad}(D) \circ \text{ad}(S) - \text{ad}(S) \circ \text{ad}(D))(C) = [\text{ad}(D), \text{ad}(S)](C) = \text{ad}([D, S])(C) = \text{ad}(0)(C) = 0$$

for all $C \in \text{End}_k(V)$. Hence $\text{ad}(D)$ and $\text{ad}(S)$ commutes. Thus $\text{ad}(T) = \text{ad}(D) + \text{ad}(S)$ is a Jordan-Chevalley decomposition. ■

3 Constructing Lie Algebras

3.1 Free Lie Algebras

Definition 3.1.1 (Free Lie Algebras)
Let R be a commutative ring. Let X be a set. Define the free Lie algebra of X over R to be a Lie algebra $L(X)$ together with a function $i : X \to L$ such that the following universal property is satisfied. For any Lie algebra A and any function $f : X \to A$, there exists a unique Lie algebra homomorphism $\widetilde{f} : L \to A$ such that the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[dr, "f"'] & L \arrow[d, dashed, "\exists! \widetilde{f}"] \\ & A \end{tikzcd}$$

Definition 3.1.2 (Hall Basis)
Let R be a commutative ring. Let X be a set. Suppose that X has a total ordering. Define the Hall basis of X to be the set $H(X)$ defined as follows.

- $X \subseteq H(X)$.
- The formal symbol $[x, y] \in H(X)$ if and only if the following are true.
  - $x < y$.
  - y is a generator or $y = [u, v]$ for $u, v \in H(X)$.

Proposition 3.1.3
Let R be a commutative ring. Let X be a set. Then the free Lie algebra of X over R is isomorphic to the R-module

$$R\langle H(X)\rangle$$

together with Lie bracket given by linearly extending the formal symbols $[x, y]$ in the Hall basis.

3.2 The Universal Enveloping Algebra

Definition 3.2.1 (The Universal Enveloping Algebra)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the universal enveloping algebra of L to be the quotient

$$U(L) = \frac{T(L)}{I}$$

where $T(L)$ is the tensor algebra of L considered as an R-module, and

$$I = \langle [x, y] - (xy - yx) \mid x, y \in L\rangle$$

Lemma 3.2.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then the following are true.

- $U(L)$ is a Lie algebra over R.
- The inclusion $L \to U(L)$ is a Lie algebra homomorphism.

Theorem 3.2.3 (Poincare-Birkhoff-Witt)
Let R be a commutative ring. Let L be a Lie algebra over R. Then the following are true.

- If L is a free R-module, then $U(L)$ is a free R-module.
- If $\{e_i \mid i \in I \subseteq \mathbb{N}\}$ is a basis of L, then $\{e_{i_1} \cdots e_{i_k} \mid i_j \in I, i_1 < \cdots < i_k\}$ is a basis for $U(L)$.

3.3 Derivations

Definition 3.3.1 (The Lie Subalgebra of All Derivations)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the Lie subalgebra of derivations to be the set

$$\text{Der}_R(L) = \{\phi \in \mathfrak{gl}_R(L) \mid \phi \text{ is a derivation } \}$$

together with Lie algebra structure inherited from $\mathfrak{gl}_R(L)$.

Lemma 3.3.2
Let R be a commutative ring. Let L be a Lie algebra over R. Let R be a commutative ring. Then $\text{Der}_R(M)$ is a Lie sub-algebra of $\mathfrak{gl}(V)$.

Proof We know that $\text{Der}_R(M)$ is an R-submodule of $\mathfrak{gl}_R(M)$. Let $f_1, f_2 \in \text{Der}_R(M)$. For any $a, b \in M$ we have

$$\begin{aligned}
[f_1, f_2](ab) &= f_1(f_2(ab)) - f_2(f_1(ab)) \\
&= f_1(f_2(a)b + af_2(b)) - f_2(f_1(a)b + af_1(b)) \\
&= f_1(f_2(a))b + f_2(a)f_1(b) + f_1(a)f_2(b) + af_1(f_2(b)) \\
&\quad - f_2(f_1(a))b - f_1(a)f_2(b) - f_2(a)f_1(b) - af_2(f_1(b)) \\
&= f_1(f_2(a))b + af_1(f_2(b)) - f_2(f_1(a))b - af_2(f_1(b)) \\
&= a[f_1, f_2](b) + [f_1, f_2](a)b
\end{aligned}$$

so that the bracket operator is closed. ∎

Lemma 3.3.3
Let R be a commutative ring. Let L be a Lie algebra over R. Then $\text{ad}(L)$ is an ideal of $\text{Der}_R(L)$.

Proof Let $x \in L$. I claim that $\text{ad}(x)$ is a derivation. Indeed, for $y, z \in L$, we have

$$\begin{aligned}
\text{ad}(x)([y, z]) &= [x, [y, z]] \\
&= -[[y, z], x] \\
&= [[z, x], y] + [[x, y], z] \\
&= -[\text{ad}(x)z, y] + [\text{ad}(x)(y), z] \\
&= [\text{ad}(x)(y), z] + [y, \text{ad}(x)(z)]
\end{aligned}$$

Thus $\text{ad}(L) \subseteq \text{Der}_R(L)$.

Let $x \in L$ and $\phi \in \text{Der}_{\mathbb{F}}(L)$. It remains to show that $[\phi, \text{ad}(x)]$ is the adjoint homomorphism of some element. We compute that

$$\begin{aligned}
[\phi, \text{ad}(x)](y) &= \phi(\text{ad}(x)(y)) - \text{ad}(x)(\phi(y)) \\
&= \phi([x, y]) - [x, \phi(y)] \\
&= [x, \phi(y)] + [\phi(x), y] - [x, \phi(y)] \\
&= [\phi(x), y] \\
&= \text{ad}(\phi(x))(y)
\end{aligned}$$

for any $y \in L$. Hence we are done. ∎

4 Types of Lie Algebras

4.1 Abelian Lie Algebras

Lie algebras that are Abelian are the simplest Lie algebra there is to study.

Definition 4.1.1 (Abelian Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. We say that L is abelian if

$$[x, y] = 0$$

for all $x, y \in L$.

Lemma 4.1.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then $Z(L)$ is abelian.

Proof True by definition of $Z(L)$. ■

Lemma 4.1.3
Let R be a commutative ring. Let L be a Lie algebra over R. Let I be an ideal of L. Then L/I is abelian if and only if

$$[L, L] \subseteq I$$

Proof Let L/I be abelian. Let $v, w \in L$. Since L/I is abelian, we have that $[v + I, w + I] = I$. But $[v + I, w + I] = [v, w] + I$ implies that $[v, w] \in I$. Conversely, suppose that $[L, L] \subseteq I$. Then for any $v + I, w + I \in L/I$, $[v + I, w + I] = [v, w] + I = I$. Hence L/I is abelian. ■

We can think of this as saying $I = [L, L]$ is the smallest ideal of L for which L/I is abelian.

4.2 Nilpotent Lie Algebras

Definition 4.2.1 (Lower Central Series)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the lower central series $L^0, L^1, \dots, L^n, \dots$ as follows.

- For $n = 0$, define $L^0 = L$
- For $n \in \mathbb{N} \setminus \{0\}$, define

$$L^n = [L, L^{n-1}]$$

Lemma 4.2.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then the following are true.

- For all $n \in \mathbb{N}$, $L^{n+1} \subseteq L^n$.
- For all $n \in \mathbb{N}$, L^n is an ideal of L.

Proof When $n = 0$, both statements are clearly true. Suppose that they are true for some $k \in \mathbb{N}$. Then by lemma 1.2.5, $L^{k+1} = [L, L^k]$ is an ideal of L, and therefore $L^{k+1} \subseteq L^k$. By induction, both statements are true for all $n \in \mathbb{N}$. ■

Lemma 4.2.3
Let R be a commutative ring. Let L_1, L_2 be Lie algebras over R. Let $\phi : L_1 \to L_2$ be a Lie algebra homomorphism. Then

$$\phi(L^k) = (\phi(L))^k$$

for all $k \in \mathbb{N}$.

Proof We prove by induction. The base case $k = 0$ is clear. Suppose that $\phi(L^k) = (\phi(L))^k$. Then we have that

$$\begin{aligned}\phi(L^{k+1}) &= \phi([L, L^k]) \\ &= [\phi(L), \phi(L^k)] \\ &= [\phi(L), \phi(L)^k] \\ &= (\phi(L))^{k+1}\end{aligned}$$

By induction, we conclude. ■

Definition 4.2.4 (Nilpotent Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. We say that L is nilpotent if there exists $n \in \mathbb{N}$ such that

$$L^n = 0$$

Lemma 4.2.5
Let R be a commutative ring. Let L be a Lie algebra over R. If L is abelian, then L is nilpotent.

Proof Let L be abelian. Then $L^1 = [L, L] = 0$. ■

4.3 Soluble Lie Algebras

Definition 4.3.1 (Derived Series)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the derived series $L^{(n)}$ of L to be the sequence recursively defined as follows.

- For $n = 0$, define $L^{(0)} = L$
- When $n \in \mathbb{N} \setminus \{0\}$, define

$$L^{(n)} = [L^{(n-1)}, L^{(n-1)}]$$

Lemma 4.3.2
Let R be a commutative ring. Let L be a Lie algebra over R. Then the following are true.

- For all $n \in \mathbb{N}$, $L^{(n+1)} \subseteq L^{(n)}$.
- For all $n \in \mathbb{N}$, $L^{(n)}$ is an ideal of L.

Proof Both are clear when $n = 0$. Suppose that they are true for some $k \in \mathbb{N}$. Then by lemma 1.2.5, $L^{(k+1)} = [L^{(k)}, L^{(k)}]$ is an ideal of L, and hence $L^{(k+1)} \subseteq L^{(k)}$. By induction, both cases are true for all $n \in \mathbb{N}$. ■

Lemma 4.3.3
Let R be a commutative ring. Let L_1, L_2 be Lie algebras over R. Let $\phi : L_1 \to L_2$ be a Lie algebra homomorphism. Then

$$\phi(L_1^{(k)}) = \phi(L_1)^{(k)}$$

Proof When $k = 0$, the lemma is trivial. Assume that it is true for some k. Then we have

$$\phi(L^{(k+1)}) = \phi([L^{(k)}, L^{(k)}]) = [\phi(L^{(k)}), \phi(L^{(k)})] = [\phi(L)^{(k)}, \phi(L)^{(k)}] = \phi(L)^{(k+1)}$$

By induction we conclude. ■

Definition 4.3.4 (Soluble Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. We say that L is soluble if there exists $n \in \mathbb{N}$ such that

$$L^{(n)} = 0$$

Lemma 4.3.5
Let R be a commutative ring. Let L be a Lie algebra over R. Then the following are true.

- If L is abelian, then L is soluble.
- If L is nilpotent, then L is soluble.

Proof Let L be abelian. Then $L^{(1)} = [L, L] = 0$.

I claim that $L^{(n)} \subseteq L^n$. When $n = 0$, the case is trivial. Suppose that $L^{(k)} \subseteq L^k$ for some $k \in \mathbb{N}$. Then we have

$$L^{(k+1)} = [L^{(k)}, L^{(k)}] \subseteq [L, L^{(k)}] \subseteq [L, L^k] = L^{k+1}$$

By induction, we conclude that $L^{(n)} = L^n$ for all $n \in \mathbb{N}$. If L is nilpotent, then $L^n = 0$ for some n so that $L^{(n)} = 0$. Then L is soluble. ■

Proposition 4.3.6
Let R be a commutative ring. Let L be a Lie algebra over R. Let I and J be ideals of L. Then the following are true.

- Let $\phi : L \to K$ be a Lie algebra homomorphism. If L is soluble then $\phi(L)$ is soluble.
- Let M be a Lie subalgebra of L. If L is soluble, then M is soluble.
- If L is soluble, then L/I is soluble.
- If I and L/I are soluble, then L is soluble.
- If I and J are soluble, then $I + J$ is soluble.

Proof

- Suppose that $L^{(n)} = 0$ for some n. Then $0 = \phi(L^{(n)}) = \phi(L)^{(n)}$ so we are done.
- When $k = 0$, we have $M^{(0)} = M \subseteq L = L^{(0)}$. Assume that $M^{(k)} \subseteq L^{(k)}$ for some $k \in \mathbb{N}$. Then we have
$$M^{(k+1)} = [M^{(k)}, M^{(k)}] \subseteq [L^{(k)}, L^{(k)}] = L^{(k+1)}$$
So by induction we have $M^{(n)} \subseteq L^{(n)}$ for all n. If L is soluble, then $L^{(n)}) = 0$ for some n so that $M^{(n)} \subseteq L^{(n)} = 0$.
- Suppose that $L^{(n)} = 0$ for some $n \in \mathbb{N}$. Let $\pi : L \to L/I$ be the projection map. Then we have
$$0 = \pi(L^{(n)}) = (L/I)^{(n)}$$
so we are done.
- Suppose that $(L/I)^{(n)} = 0$ for some $n \in \mathbb{N}$. Let $\pi : L \to L/I$ be the projection map. Since π is surjective, we have
$$\pi(L^{(n)}) = (L/I)^{(n)} = 0$$
This means that $L^{(n)} \subseteq I$. We know that I being soluble means that $I^{(k)} = 0$ for some k. Therefore we have
$$L^{(n+k)} \subseteq I^{(k)} = 0$$
so that L is soluble.
- By the third isomorphism theorem, we have a Lie algebra isomorphism $\frac{I+J}{J} \cong \frac{I}{I \cap J}$. Since I is soluble, $\frac{I}{I \cap J}$ is soluble by the above. Hence $\frac{I+J}{J}$ is soluble. Since J is also soluble, from the above we conclude that $I + J$ is soluble.

■

4.4 Semisimple Lie Algebras

Lemma 4.4.1
Let R be a commutative ring. Let L be a Lie algebra over R. Then there exists a unique soluble ideal S of L such that for any soluble ideal $J \subseteq L$, we have $J \subseteq S$.

Proof Every Lie algebra has a soluble ideal $Z(L)$. Hence the set of all soluble ideals of L is non-empty. Then it has a maximal element S. For any other soluble ideal I, $S + I$ is soluble and is either $S + I = L$ or $S + I = S$. Hence S is the unique maximal soluble ideal containing all other soluble ideals. ■

Definition 4.4.2 (Radical Ideals)
Let R be a commutative ring. Let L be a Lie algebra over R. Define the radical ideal $\text{rad}(L) \subseteq L$ of L to be the unique soluble ideal of L that contains all other soluble ideals.

Definition 4.4.3 (Semisimple Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. We say that L is semisimple if

$$\text{rad}(L) = \{0\}$$

Lemma 4.4.4
Let R be a commutative ring. Let L be a Lie algebra over R. Then $L/\text{rad}(L)$ is semisimple.

Proof Let K be a soluble ideal of $L/\text{rad}(L)$. By the correspondence theorem, there exists an ideal I of L such that $\text{rad}(L) \subseteq I$ and $K = I/\text{rad}(L)$. Since $\text{rad}(L)$ and K are soluble, we conclude that I is soluble. Hence $I \subseteq \text{rad}(L)$. We conclude that $I = \text{rad}(L)$. Hence $K = \{0\}$. Thus $L/\text{rad}(L)$ is semisimple. ■

Lemma 4.4.5
Let R be a commutative ring. Let L be a Lie algebra over R. Then L is not semisimple if and only if L contains a non-trivial abelian ideal.

Proof If L is not semisimple, then $\text{rad}(L) \neq \{0\}$ is a non-trivial soluble ideal. This means that there exists a smallest $n \in \mathbb{N}$ such that $\text{rad}(L)^{(n)} = 0$. But this is the same as saying that

$$[\text{rad}(L)^{(n-1)}, \text{rad}(L)^{(n-1)}] = \text{rad}(L)^{(n)} = 0$$

This means that $\text{rad}(L)^{(n-1)}$ is an abelian ideal. In particular, it is non-trivial since n is the smallest number for which $\text{rad}(L)^{(n)}$ is zero.

If L contains a non-trivial abelian ideal I, then by lmm 2.2.4 we have that I is soluble. Hence $I \subseteq \text{rad}(L)$ and $\text{rad}(L)$ is non-zero. Hence L is not semisimple. ■

4.5 Simple Lie Algebras

Definition 4.5.1 (Simple Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. We say that L is simple if L is non-abelian and has no proper non-zero ideals.

Lemma 4.5.2
Let R be a commutative ring. Let L be a Lie algebra over R. If L is simple, then L is semisimple.

Proof Since L has no non-trivial proper ideals, $\text{rad}(L) = \{0\}$ or L. If $\text{rad}(L) = L$, then L is abelian and hence a contradiction. ■

5 Modules over Lie Algebras

5.1 Modules over Lie Algebras

Definition 5.1.1 (Modules over Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. An L-module consists of an R-module M together with an action

$$\cdot : L \times M \to M$$

such that the following are true.

- Linearity in L: For all $a, b \in R$ and $x, y \in L$, we have that
$$(ax + by) \cdot v = a(x \cdot v) + b(y \cdot v)$$
for all $v \in M$.
- Linearity in V: For all $x \in L$, we have that
$$x \cdot (av + bw) = a(x \cdot v) + b(x \cdot w)$$
for all $a, b \in k$ and $v, w \in M$.
- Commutes with the Lie bracket: $[x, y] \cdot v = x \cdot (y \cdot v) - y \cdot (x \cdot v)$.

Definition 5.1.2 (Submodule of Modules over Lie Algebras)
Let R be a commutative ring. Let L be a Lie algebra over R. Let M be an L-module. Let $N \subseteq M$ be a subset. We say that N is a Lie submodule of M if N is a module over L.

Definition 5.1.3 (L-Module Homomorphisms)
Let R be a commutative ring. Let L be a Lie algebra over R. Let M, N be L-modules. Let $f : M \to N$ be an R-module homomorphism. We say that φ is an L-module homomorphism if

$$\phi(x \cdot v) = x \cdot \phi(v)$$

for all $x \in L$ and $v \in M$.

Definition 5.1.4
Let R be a commutative ring. Let L be a Lie algebra over R. Let M, N be L-modules. Define the R-module of L-module homomorphisms from M to N to be the R-module

$$\text{Hom}_L(M, N) = \{f : M \to N \mid f \text{ is an } L\text{-module homomorphism }\}$$

Lemma 5.1.5
Let R be a commutative ring. Let L be a Lie algebra over R. Let M, N be L-modules. Then $\text{Hom}_L(M, N)$ is an R-submodule of $\text{Hom}_R(M, N)$.

Definition 5.1.6 (L-Module Isomorphisms)
Let R be a commutative ring. Let L be a Lie algebra over R. Let M, N be L-module. Let $f : M \to N$ be an R-module homomorphism.We say that f is an L-module isomorphism if there exists an L-module homomorphism $g : N \to M$ such that

$$g \circ f = \text{id}_M \quad \text{and} \quad f \circ g = \text{id}_N$$

5.2 The Invariants of a Module

Definition 5.2.1 (The Invariants of a Modules)
Let R be a commutative ring. Let L be a Lie algebra over R. Let M be an L-module. Define the invariants of M to be

$$M^L = \{m \in M \mid xm = 0 \text{ for all } x \in L\}$$

Lemma 5.2.2
Let R be a commutative ring. Let L be a Lie algebra over R. Let M be an L-module. Then M^L is an L-submodule of M.

Lemma 5.2.3
Let R be a commutative ring. Let L be a Lie algebra over R. Let M, N be L-module. Then there is an R-module homomorphism

$$\text{Hom}_L(M, N) \cong \text{Hom}_R(M, N)^L$$

The above isomorphism of R-modules give $\text{Hom}_L(M, N)$ the structure of an L-module.

5.3 Reducibility of Modules over Lie Algebras

Definition 5.3.1 (Direct Sum of Modules over Lie Algebras)
Let L be a Lie algebra. Let V be an L-module. Let U and W be vector subspaces of V. We say that V is the direct sum of U and W if

$$V = U \oplus W$$

as vector spaces and U and W are L-submodules of V.

Definition 5.3.2 (Irreducible Modules over Lie Algebras)
Let L be a Lie algebra. Let V be an L-module. We say that V is irreducible (simple) if V has no proper non-trivial L-submodules.

Definition 5.3.3 (Completely Reducible)
Let L be a Lie algebra. Let V be an L-module. We say that V is completely reducible if for all L-submodules W, there exists an L-submodule U of V such that

$$V = W \oplus U$$

Simple modules are thus vacuously completely reducible.